

Loan Portfolio Risk & Performance Analytics

An end-to-end analytics case study — from raw loan data to an interactive Power BI dashboard, built to support credit risk, pricing, and portfolio decisions.

Background & Business Problem

Background

LendingClub is a peer-to-peer (P2P) lending platform that connects borrowers seeking personal loans with investors willing to fund them. Between 2007 and 2018, LendingClub issued over 2.2 million loans totalling billions of dollars across the United States — making it the world's largest P2P lending platform at its peak. Unlike traditional banks, LendingClub operates without branches, relying entirely on data-driven credit assessment to approve loans, assign interest rates, and manage risk.

Business Problem

In a lending business, every loan that defaults represents a direct financial loss — the principal is written off as a charge-off and reduces the company's profitability. LendingClub's core challenge is: how do you lend to as many borrowers as possible to grow revenue, while keeping defaults low enough to protect margins? This tension between growth and risk sits at the heart of every credit business.

As a Data Analyst, the business expects answers to four critical questions:

1. Which borrower segments carry the highest default risk, and what patterns — loan purpose, credit score, debt-to-income ratio, loan grade — predict default?
2. Is the loan portfolio growing, and which loan grades, purposes, and loan terms are driving that growth?
3. Do riskier borrowers pay higher interest rates, and is that gap large enough to plausibly offset their higher default rate?
4. How many loans and how much funded value is being lost to default, relative to the size of the overall portfolio?

Objective

This project builds an end-to-end analytics solution — from raw data to interactive dashboard — that gives LendingClub's credit, pricing, and finance teams a single source of truth to monitor portfolio health, identify high-risk borrower profiles, track origination trends, and measure credit loss. The goal is not just to visualise data but to produce actionable insights that directly support lending decisions.

Dataset

The dataset contains **890,000+** real loan records sourced from LendingClub's SEC filings (2007–2018), with **74+ columns** covering borrower financials (income, DTI, FICO score, credit utilisation), loan attributes (grade, interest rate, purpose, term), and repayment outcomes (loan status, charge-off amount, recoveries).

Data Cleaning & Transformation

1 Data Cleaning (Python — NumPy & Pandas)

The raw LendingClub dataset (`accepted_2007_to_2018Q4.csv`) contains millions of loan records across a wide range of fields. To keep the analysis focused and memory-efficient, only 26 relevant columns were loaded at read time — covering loan terms, borrower attributes, credit history, and loan outcome.

- 1 Filtering to completed loans** — the dataset includes loans still in progress (e.g. Current, Late), which have no resolved outcome. Data was filtered to only *Fully Paid* and *Charged Off* loans — the two statuses with a definitive result.
- 2 Handling missing values** — rather than dropping any row with a null value anywhere, missing values were removed only where they occurred in four critical columns: `emp_length` , `dti` , `revol_util` , `annual_inc` .
- 3 Data type conversion** — `issue_d` and `earliest_cr_line` were converted to proper datetime fields, enabling downstream time-based analysis.
- 4 Feature engineering** — `avg_fico` (midpoint of the FICO range) and `default_flag` (binary target: 1 = Charged Off, 0 = Fully Paid) were derived for the risk analysis.

The cleaned dataset was exported to `cleaned_data.csv` for loading into Power BI.

2 Data Loading & Transformation (Power Query)

The cleaned CSV was loaded into Power BI, where Power Query handled a second layer of transformation — turning continuous numeric fields into categorical bands for segment-level risk analysis.

Column	Based on	Bands
<code>dti_band</code>	Debt-to-income	Low (<10), Medium (10–20), High (20–30), Very High (30+)
<code>fico_band</code>	FICO score	Poor (<650), Fair (650–699), Good (700–749), Excellent (750+)
<code>Income_tier</code>	Annual income	Low (<40K), Middle (40K–80K), Upper-Mid (80K–120K), High (120K+)
<code>Loan_bucket</code>	Loan amount	Small (<5K), Medium (5K–15K), Large (15K–25K), Very Large (25K+)
<code>revol_util_band</code>	Revolving utilisation	Low (<25%), Medium (25–50%), High (50–75%), Very High (75%+)

Year, Quarter, and Month columns were also derived from the issue date, enabling the time-based slicing used on the Portfolio Growth page.

Date Table & DAX Measures

3 Date Table (Power Query)

To support time-intelligence calculations like Year-over-Year growth and running totals, a dedicated Date table was built rather than relying on the dates embedded in the loans table — Power BI's time-intelligence functions need a continuous, gap-free calendar to work correctly.

The table was generated using `List.Dates`, creating one row per calendar day from January 1, 2007 to December 31, 2018:

```
= List.Dates(#date(2007,1,1),  
            Duration.Days(#date(2018,12,31) - #date(2007,1,1)) + 1,  
            #duration(1,0,0,0))
```

Four supporting columns were added — **Year**, **Quarter**, **Month**, and **MonthName** — and the table was linked to the loans table via `issue_d` in Model view.

4 Measures (DAX)

A set of DAX measures was built on the transformed data model to power every KPI and comparison on the dashboard, falling into three groups:

Core Portfolio Measures

- `Total Loans` — count of all loans
- `Total Funded Amount` — total dollar value lent
- `Total Defaults` — count of charged-off loans
- `Default Percentage` — portfolio-wide default rate
- `Average Loan Amount` — funded amount divided by loan count

Growth Measure

`YoY Funded Growth %` compares the current year's funded amount to the prior year using `CALCULATE` and `FILTER` over the Date table's Year column, returning the percentage change via `DIVIDE`.

Risk Profile Comparison Measures

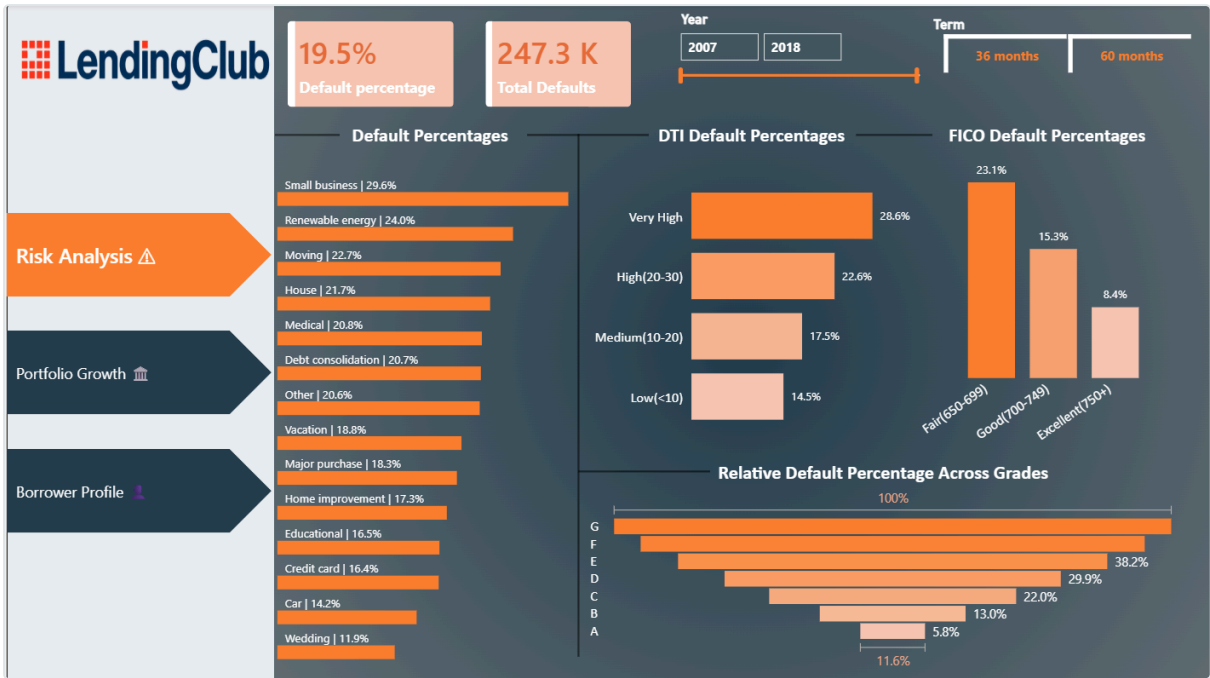
A parallel set of "Charged Off vs. Fully Paid" measures — covering average DTI, FICO, income, interest rate, revolving balance, and revolving utilisation — isolates each metric by loan outcome using `CALCULATE` with a `loan_status` filter. This powers the borrower risk-profile comparisons on the dashboard.

Visualization (Power BI)

The final dashboard is structured across three pages, each built to answer a specific set of the business questions raised earlier.

5 Risk Analysis

This page examines default risk across the dimensions that drive it — loan purpose, credit utilisation (DTI), credit score (FICO), and loan grade — answering which borrower segments carry the highest default risk.



247.3K

TOTAL DEFAULTS

19.5%

DEFAULT PERCENTAGE

Default Percentage by Purpose surfaces which borrowing reasons carry disproportionate risk. Small business loans top the list at a 29.6% default rate — more than 2.5× the rate for the safest category (wedding loans, 11.9%).

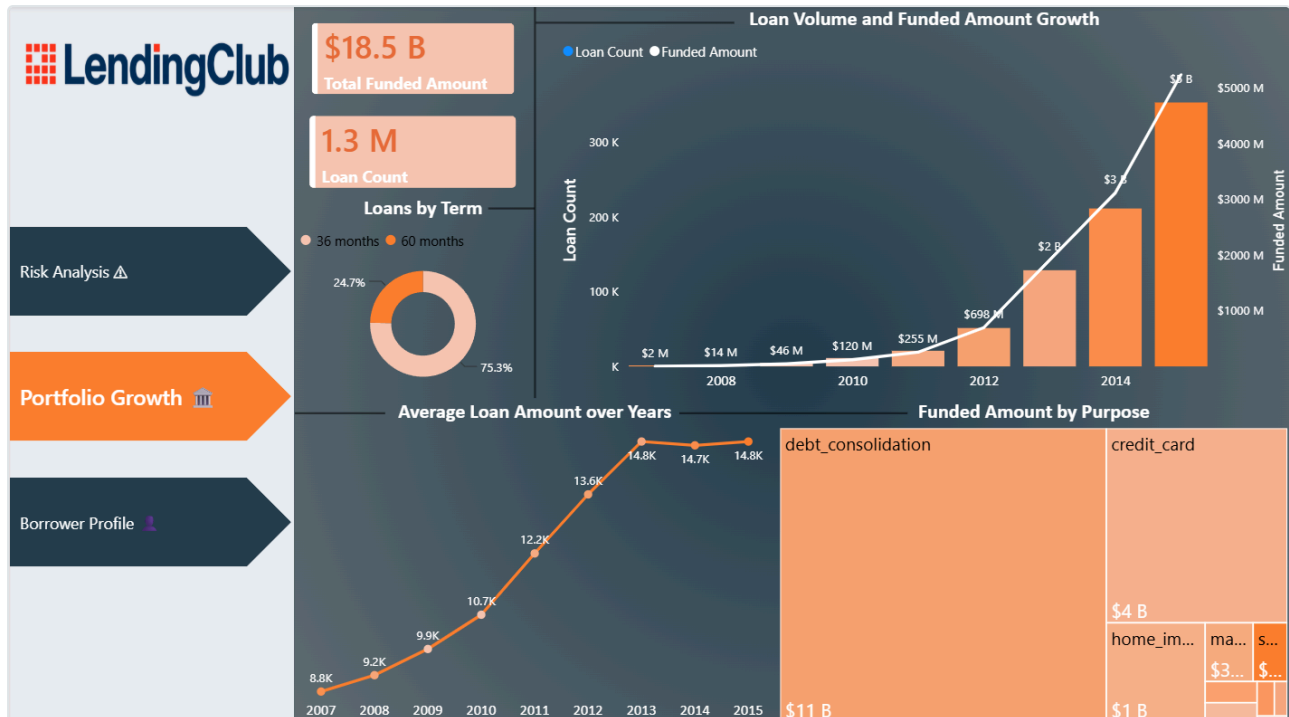
DTI Default Percentages shows default rate climbing steadily with debt-to-income ratio, from 14.5% (Low) to 28.6% (Very High) — confirming DTI as a strong, monotonic predictor of default.

FICO Default Percentages shows the inverse pattern: 23.1% for Fair credit down to 8.4% for Excellent — nearly a 3× spread.

Relative Default Percentage Across Grades shows a near-linear escalation in risk from Grade A (11.6% relative) to Grade G (100% relative), showing LendingClub's grading system tracks closely with realised outcomes.

Portfolio Growth

This page steps back from individual risk factors to track the company's growth trajectory over its lifetime — is the portfolio growing, and what's driving it.



\$18.5B

TOTAL FUNDED AMOUNT

1.3M

LOAN COUNT

Loan Volume & Funded Amount Growth is the centrepiece of the page, functioning like a revenue-growth chart for the business — from a near-zero base in 2007–08 to \$255M by 2012, then a sharp acceleration to roughly \$5B annually by 2014–15.

Loans by Term shows 75.3% of loans are 36-month terms versus 24.7% at 60 months, indicating the book is weighted toward shorter-duration lending.

Average Loan Amount over Years rose from \$8.8K (2007) to a plateau around \$14.7–\$14.8K from 2013 onward — growth wasn't just more loans, but larger average loan sizes.

Funded Amount by Purpose shows debt consolidation dominating at \$11B — more than double the next-largest category, credit cards (\$4B).

Borrower Profile

This page isolates the borrower-level differences between loans that were repaid and loans that defaulted — do riskier borrowers pay proportionally higher interest, and does that premium offset their added risk.



Risk Analysis

Portfolio Growth

Borrower Profile

Fully Paid vs Charged Off

	Fully Paid	Charged Off
Average Income	\$79.2 K	\$72.4 K
Average DTI	17.7	20.0
Average FICO	700.2	689.7
Average Revolving Balance	\$16.6 K	\$15.6 K
Average Loan Interest	13%	16%
Average Utilization Ratio	51%	55%

Metric	Fully Paid	Charged Off
Average Income	\$79.2K	\$72.4K
Average DTI	17.7	20.0
Average FICO	700.2	689.7
Average Revolving Balance	\$16.6K	\$15.6K
Average Loan Interest	13%	16%
Average Utilisation Ratio	51%	55%

The pattern is consistent across every metric: borrowers who defaulted had lower income, higher debt-to-income, lower credit scores, and higher credit utilisation on average.

Key figure: charged-off borrowers paid a 3-percentage-point higher average interest rate (16% vs. 13%). Whether that premium fully offsets a 19.5% portfolio-wide default rate is addressed in the Findings section.

Design

Beyond the analysis, the dashboard's visual design was deliberately built rather than left to Power BI defaults, aiming for a cohesive, brand-aligned look.

Colour Palette

Drawn directly from LendingClub's brand identity, sourced and refined using Colorhunt.co:

■ Navy Blue #233D4D ■ Orange #FE7F2D ■ Black □ White

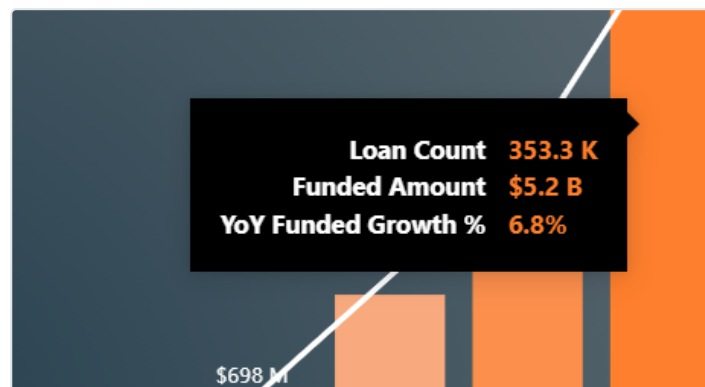
Different shades of orange are used via conditional formatting to represent variation within a single metric — e.g. severity gradients on the Risk Analysis charts.

Canvas Layout & Navigation

The page layout — a white navigation panel against a navy gradient background — was first prototyped in Google Slides, then rebuilt natively in Power BI. Navigation is built using Power BI's Navigator feature: the active page is highlighted in orange, inactive pages shown in navy, giving a clear, brand-consistent wayfinding cue.

Tooltips

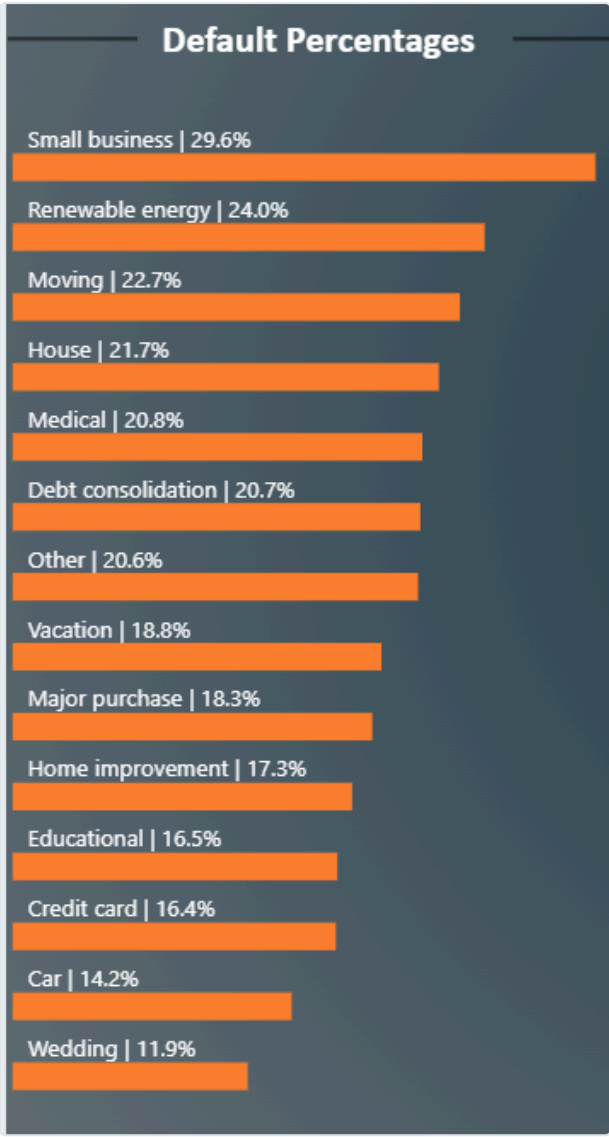
Default Power BI tooltips were replaced with a custom design — black background, orange values, white labels — keeping tooltips legible and visually consistent with the rest of the dashboard.



Notable Chart: Default Percentage by Purpose

Standard clustered bar charts place data labels at the bar's end, which becomes cramped with 15 long category names. A custom labelling technique solved this:

- 1 A placeholder measure (`Labels Placeholder = 0`) added alongside the default percentage renders as a zero-length bar, creating clean spacing between bars without consuming chart width.
- 2 A DAX measure (`Labels Bar Chart`) dynamically builds a combined label — e.g. "*Small business* | 29.6%" — cleaning underscores and capitalising the purpose name.
- 3 This custom label is applied directly on top of each bar via Power BI's data label settings.



Findings

1. Which segments carry the highest default risk, and what predicts it?

Default risk is strongly predictable from four factors: loan purpose (Small Business at 29.6% vs. Wedding at 11.9%), DTI (14.5% Low rising to 28.6% Very High), FICO (23.1% Fair vs. 8.4% Excellent), and LendingClub's own grade (Grade A at 11.6% relative vs. Grade G at 100%). All four signals move in a consistent, monotonic pattern with risk.

2. Is the portfolio growing, and what's driving it?

The portfolio grew from a near-zero base in 2007 to \$18.5B in total funded amount and 1.3M loans by 2018, with the sharpest acceleration from 2012 onward (~\$255M to ~\$5B annually). Growth was driven both by more loans and larger average loan sizes (\$8.8K → \$14.8K). Debt consolidation is the dominant driver at \$11B — more than double the next-largest category.

3. Do riskier borrowers pay proportionally higher interest?

Yes, directionally: charged-off borrowers paid a 16% average rate versus 13% for fully-paid borrowers — a 3-point premium. However, this group also had lower average FICO (689.7 vs. 700.2), higher DTI (20.0 vs. 17.7), and lower income (\$72.4K vs. \$79.2K). Whether a 3-point premium fully compensates for that combined risk gap, given a 19.5% portfolio-wide default rate, remains an open question flagged for pricing review.

4. How much is being lost to default relative to portfolio size?

247.3K loans — 19.5% of all completed loans in the dataset — ended in charge-off, representing a substantial share of originated volume left unrecovered.

Business Recommendations

- › **Tighten underwriting on high-risk purpose categories.** Small business, renewable energy, and moving loans default well above the portfolio average — these warrant either stricter approval criteria or higher risk-based pricing.
- › **Re-examine whether the interest rate premium is sufficient.** The current 3-point gap between fully-paid and charged-off average rates may not fully price in the observed DTI, FICO, and income differences — worth a deeper repricing analysis by grade.
- › **Use DTI and FICO bands as early screening signals.** Both show a clean, monotonic relationship with default risk, making them strong candidates for automated pre-approval thresholds.
- › **Monitor concentration risk in debt consolidation.** At \$11B of \$18.5B total funded, over-reliance on a single loan purpose increases exposure if that segment's risk profile shifts.
- › **Grade-level default rates validate the existing grading system.** Grade A through G shows a near-linear risk escalation, suggesting the underlying grading model is well-calibrated and doesn't need structural rework — just continued monitoring.